Testbed Protocol for Adaptive Superdense Coding in Distributed Quantum Networks

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***Abstract*—Distributed quantum networks require robust protocols for high-capacity, noise-resilient communication between spatially separated nodes. Superdense coding, which leverages shared entanglement to transmit two classical bits per qubit, is a key primitive for benchmarking such networks. However, entanglement degradation from channel noise and decoherence poses major challenges for achieving scalable throughput in practical environments. We introduce a testbed protocol for adaptive superdense coding in distributed quantum networks (DQNs), integrating the five-qubit error correction code with a global, feedback-driven purification routine. By continuously monitoring quantum discord (QD) and entanglement of formation (EoF) via dedicated pilot pairs, our protocol dynamically tunes purification parameters to counteract amplitude and phase damping, while avoiding the loss of entangled pairs inherent to traditional schemes. Simulation results demonstrate that this hybrid quantum-classical approach enhances channel fidelity and throughput under realistic noise, providing a scalable testbed for distributed quantum error correction strategies and hybrid control in next-generation quantum networks.**

***Index Terms*—Adaptive purification, Distributed quantum networks, Error correction, Quantum discord, Entanglement of formation, Pilot pairs, Damping channels, Fidelity**

I. Introduction

Distributed quantum networks are poised to revolutionize scalable quantum computing and secure communications by connecting spatially separated quantum nodes via entanglement and quantum channels [1]. Unlike classical networks, distributed quantum networks leverage nonlocal correlations, enabling protocols such as entanglement swapping, quantum teleportation, and superdense coding across multi-node architectures [2]. Among these, superdense coding stands out as a benchmark protocol for testing high-capacity quantum information transfer and resource efficiency in realistic networked environments [3], [4]. Despite the theoretical efficiency of superdense coding, which can transmit two classical bits per qubit, its practical deployment remains challenging. A major obstacle is the noise and decoherence of quantum channels, which limit channel capacity and operational performance [5], [6], [7], [8]. Early experiments over optical fiber and free-space links have consistently reported suboptimal performance, underscoring the need for robust error mitigation strategies suited to distributed, real-world environments.

Traditional solutions, such as quantum error correction (QEC) and entanglement purification have made notable progress toward combating noise in isolated quantum systems [9]. However, distributed architectures introduce additional complexities, such as channel asymmetry, node synchronization errors, and multi-qubit noise issues that are only partially addressed by existing schemes [10]. QEC codes can protect against single-qubit errors, but often struggle with correlated errors in network scenarios. Likewise, conventional purification protocols based on local operations and classical communication (LOCC) [11], [12] achieve high-fidelity pairs by discarding many noisy ones, a practice that sharply reduces overall throughput in dense coding applications across distributed channels.

Recent advances have motivated adaptive approaches that dynamically tailor QEC and purification steps to network conditions [13], [14], [15], [16]. *However, a critical challenge remains: how to implement adaptive purification in distributed settings without sacrificing valuable entangled pairs or incurring prohibitive resource overhead.* In this work, we present a testbed protocol for adaptive superdense coding in distributed quantum networks. Our protocol combines the five-qubit perfect code for single-qubit error protection with a novel global purification step whose parameters are tuned in real time, guided by live feedback from quantum correlation metrics. Specifically, we employ *quantum discord* (QD) and *entanglement of formation* (EoF): two complementary information-theoretic quantities as real-time indicators of channel noise. QD and EoF have previously been linked to key rates in quantum cryptography and teleportation fidelity, respectively [17], [18], and here we demonstrate their predictive power for adaptive purification in dense coding. Our approach leverages dedicated "pilot pairs": entangled pairs traversing same network path but carrying no user data, to monitor the evolving noise environment. This enables each node to adjust purification parameters on the fly, optimizing throughput and fidelity without discarding any communication pairs.

By embedding this protocol within a network-oriented architecture, we provide a simulation-based testbed for benchmarking adaptive error correction, noise estimation, and hybrid quantum-classical control in distributed quantum networks [19]. This approach not only supports scalable evaluation for multi-user, multi-hop networks, as the pilot-pair overhead is amortized with system size, but also makes adaptive superdense coding accessible to researchers without direct hardware access. By democratizing DQC exploration, our open testbed fosters broader participation and accelerates innovation in error correction and hybrid control for noisy quantum networks.

Contributions. The main contributions of this work are:

1) We introduce an adaptive superdense coding protocol for

distributed quantum networks that integrates five-qubit QEC with global, feedback-driven purification based on real-time quantum correlation metrics.
2) We demonstrate that quantum discord and entanglement of formation are highly predictive of channel fidelity, enabling dynamic adjustment of purification operations without loss of channel capacity.
3) Our protocol serves as a practical testbed for benchmarking hybrid quantum-classical control and adaptive error correction strategies in realistic, noisy network scenarios.

The remainder of this paper is organized as follows. Section II provides a concise theoretical background, including the role of superdense coding and QEC in distributed networks. Section III details the adaptive protocol and testbed implementation. Section IV presents simulation results and benchmarking analyses. Section V discusses implications for scalable distributed quantum architectures, and Section VI concludes with future directions for experimental validation and protocol extensions.

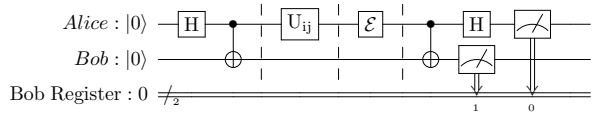


Figure 1: Superdense Coding scheme. If $\mathcal{E} = I$, Bob will receive correct bitstring ij. However, for noisy channels, $\mathcal{E} \neq I$.

II. Background and Technical Overview

This section outlines the theoretical foundations of our adaptive superdense coding protocol in distributed quantum networks. We introduce the essential components: superdense coding, quantum noise models, quantum correlations, error correction, and entanglement purification—emphasizing their roles and challenges in networked quantum systems.

A. Superdense Coding in Distributed Networks

Superdense coding enables the transmission of two classical bits with a single qubit, provided a maximally entangled Bell state is shared between sender and receiver [4]. In distributed quantum networks, this protocol serves as a benchmark for evaluating entanglement distribution, channel capacity, and network fidelity [1]. The entanglement resource (**ebit**) is essential for nonlocal information transfer across nodes. Figure 1 illustrates the protocol: Alice encodes her qubit via $U_{ij} = Z^i X^j$ and transmits her qubit through the network to Bob, who decodes by Bell measurement. The capacity reaches two bits per qubit in ideal conditions, which reduces due to noise and decoherence in practice [9].

B. Quantum Noise Channels in Networks

Quantum channels in real-world networks are subject to various noise processes, typically modeled as completely positive, trace-preserving (CPTP) maps [20]. The dominant mechanisms in photonic links are amplitude damping (energy loss, e.g., photon loss) and phase damping (loss of coherence) [21], [22], described by Kraus operators:

$$\mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger, \quad \sum_i K_i^\dagger K_i = I. \tag{1}$$

The combined effects of such noise present a major challenge for reliable entanglement distribution in distributed architectures.

Let ρ denote the two-qubit density matrix of the noisy channel output, and let σ be the ideal Bell state. To quantify the effect of noise, we use the fidelity, defined as

$$F(\rho, \sigma) = \left(\mathrm{Tr} \sqrt{\sqrt{\rho}\, \sigma\, \sqrt{\rho}} \right)^2. \tag{2}$$

C. Quantum Correlations as Network Noise Indicators

Quantum correlations are fundamental to the performance of distributed protocols [23], [24]. Two quantities are central to our approach:

Entanglement of Formation (EoF): For a bipartite state ρ, EoF quantifies minimum entanglement required [18] to prepare ρ, directly assessing entanglement quality in network.
Quantum Discord (QD): QD captures quantum correlations beyond entanglement, remaining nonzero even for separable states, making it useful for detecting subtle noise effects [24]. In our protocol, both EoF and QD are monitored using network "pilot pairs" for real-time adaptation to channel conditions.

D. Quantum Error Correction in Distributed Settings

Quantum error correction (QEC) is crucial for scalable, reliable quantum networking, allowing logical qubits to be encoded over multiple physical qubits and errors detected via stabilizer measurements [20], [25]. The quantum Hamming bound specifies the maximal number of errors that a code of given length can correct. The five-qubit code, which saturates the quantum Hamming bound, corrects all single-qubit errors and is well-suited for distributed and experimental implementations [26]:

$$\begin{aligned} g_1 &= XZZXI, \quad g_2 = IXZZX, \\ g_3 &= XIXZZ, \quad g_4 = ZXIXZ. \end{aligned} \tag{3}$$

Distributed settings introduce challenges such as correlated errors, asymmetric noise, and synchronization, motivating adaptive QEC protocols [10].

E. Entanglement Purification and Its Limitations

Entanglement distillation (purification) enhances fidelity by probabilistically converting multiple noisy pairs into fewer, high-quality entangled states using local operations and classical communication (LOCC) [11], [23]. Standard protocols (BBPSSW, DEJMPS [11], [12]). Pair discarding refers to the removal of noisy entangled pairs during standard purification protocols, thereby reducing throughput.

Our protocol addresses these limitations by integrating QEC with adaptive, non-discarding purification, guided dynamically by quantum correlations, and implemented as a testbed for benchmarking error correction strategies in distributed quantum networks.

III. METHODOLOGY

A. Quantum Communication Channel Modeling

We model each network link as a quantum channel subject to both amplitude damping (AD) and phase damping (PD). The amplitude damping channel, which represents energy dissipation (e.g., photon loss), is described by Kraus operators:

$$A_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix} \tag{4}$$

where p is the damping probability.

The phase damping channel, responsible for pure dephasing without energy loss, uses Kraus operators:

$$P_0 = \sqrt{1-q}\, I, \quad P_1 = \sqrt{q} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_2 = \sqrt{q} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{5}$$

where q is the dephasing probability.

To capture the combined effect, we use a composite channel acting on the state ρ:

$$\mathcal{E}(\rho) = \frac{1}{2} \left(AD(PD(\rho)) + PD(AD(\rho)) \right) \tag{6}$$

This model simulates realistic entanglement degradation affecting each link in a distributed quantum network.

B. Entropy Metrics Calculation

To monitor entanglement and quantum correlations in the noisy channel, we compute two metrics: *Entanglement of Formation* (EoF) and *Quantum Discord* (QD). EoF quantifies the amount of entanglement, while QD captures all quantum correlations, including those beyond entanglement.

Quantum Discord: QD is defined as the difference between quantum and classical mutual information,

$$\delta(A:B) = I(A:B) - J(A:B) \tag{7}$$

where the quantum mutual information is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho) \tag{8}$$

with $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ the von Neumann entropy, and ρ_A (ρ_B) the reduced density matrix of subsystem A (B).

The classical mutual information $J(A:B)$ is computed using Algorithm 1, which numerically optimizes over all projective measurements on subsystem A, parameterized by θ and ϕ. For each measurement setting, the algorithm defines projectors Π_0 and Π_1, computes outcome probabilities p_k, and conditional states $\rho_{B|k}$, and then minimizes the average conditional entropy over all measurement settings. The final classical mutual information is:

$$J(A:B) = S(\rho_B) - \min_{\theta,\phi} \sum_k p_k S(\rho_{B|k}) \tag{9}$$

where $S(\cdot)$ is the von Neumann entropy.

Entanglement of Formation: EoF is computed in two steps, using the concurrence as an intermediate quantity. The concurrence for a two-qubit state ρ is

$$C = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4) \tag{10}$$

where the λ_i are square roots of the eigenvalues of $R = \rho(Y \otimes Y)\rho^*(Y \otimes Y)$, in descending order, with Pauli-Y matrix.

Algorithm 1 Classical Mutual Information

Input: Two-qubit density matrix ρ
Output: Classical mutual information $J(A:B)$

function CMI(ρ)
  $\rho_A \leftarrow \mathrm{Tr}_B(\rho)$
  $\rho_B \leftarrow \mathrm{Tr}_A(\rho)$
  $f \leftarrow 0$
  for $\theta \in [0,\pi], \phi \in [0, 2\pi]$ **do**
    $\Pi_0 \leftarrow \begin{pmatrix} \cos^2(\theta/2) & \frac{\sin\theta}{2} e^{-i\phi} \\ \frac{\sin\theta}{2} e^{i\phi} & \sin^2(\theta/2) \end{pmatrix}$
    $\Pi_1 \leftarrow \begin{pmatrix} \sin^2(\theta/2) & -\frac{\sin\theta}{2} e^{-i\phi} \\ -\frac{\sin\theta}{2} e^{i\phi} & \cos^2(\theta/2) \end{pmatrix}$
    $H \leftarrow 0$
    for $k \in \{0,1\}$ **do**
      $p_k \leftarrow \mathrm{Tr}((\Pi_k \otimes I)\rho(\Pi_k \otimes I)^\dagger)$
      if $p_k > 0$ **then**
        $\rho_k \leftarrow \mathrm{Tr}_A \left(\frac{(\Pi_k \otimes I)\rho(\Pi_k \otimes I)^\dagger}{p_k} \right)$
        $H \leftarrow H + p_k \cdot S(\rho_k)$
      end if
    end for
    $C \leftarrow S(\rho_B) - H$
    if $C > f$ **then**
      $f \leftarrow C$
    end if
  end for
  $J \leftarrow f$
  return J
end function

Algorithm 2 Concurrence

Input: Two-qubit density matrix ρ
Output: Concurrence C (entanglement measure)

function CONCURRENCE(ρ)
  $Y \leftarrow$ Pauli Y matrix
  $R \leftarrow \rho \times (Y \otimes Y) \times \rho^* \times (Y \otimes Y)$
  eigenvalues $\leftarrow$ eigenvalues of R
  Eig $\leftarrow$ sort eigenvalues descending
  $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \leftarrow \sqrt{\text{Eig}[0]}, \sqrt{\text{Eig}[1]}, \sqrt{\text{Eig}[2]}, \sqrt{\text{Eig}[3]}$
  $C \leftarrow \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4)$
  return C
end function

Given concurrence C, EoF is computed as binary entropy so that $E_F(\rho) = h(x)$. The full procedure is outlined in Algorithm 3:

$$h(x) = -x \log_2 x - (1-x)\log_2(1-x), \quad x = \frac{1 + \sqrt{1 - C^2}}{2} \tag{11}$$

Algorithm 3 Entanglement of Formation

Input: Two-qubit density matrix ρ
Output: Entanglement of formation $E_F(\rho)$

function ENTANGLEMENTOFFORMATION(ρ)
  $C \leftarrow$ Concurrence(ρ)
  if $C = 0$ **then**
    return 0
  else
    $x \leftarrow (1 + \sqrt{1 - C^2})/2$
    $h \leftarrow -x \log_2(x) - (1-x)\log_2(1-x)$
    return h
  end if
end function

In summary, these procedures yield quantitative metrics for entanglement and general quantum correlations under realistic noisy conditions, as required for adaptive channel characterization and protocol optimization. Finally, fidelity with respect to the ideal Bell state is used as an operational performance metric:

$$F = (\mathrm{Tr}\sqrt{\rho})^2 \tag{12}$$

where ρ is the noisy output state after channel evolution.

IV. Quantum Correlations as Metrics

To monitor and adapt to channel noise in distributed quantum networks, we calculate both *Quantum Discord* (QD) and *Entanglement of Formation* (EoF) as diagnostic metrics for quantum correlations. As shown in Fig. 2, while Fidelity and EoF typically degrade in tandem as noise increases, QD often remains nonzero even when EoF vanishes. This persistence indicates that nonclassical correlations can survive the loss of entanglement, a key consideration for distributed settings where noise may vary between links or over time. Furthermore, quantum states with the same fidelity can exhibit differing levels of QD and EoF, implying that they may possess varying quantum resources despite equivalent proximity to the ideal Bell state.

To quantitatively assess the relationship between these correlation measures and the effective quality of our quantum states, we perform a linear regression to determine if a combination of QD and EoF can reliably predict fidelity. As summarized in Table 1, our model achieves a strong fit ($R^2 \approx 0.98$, $MSE < 10^{-3}$), according to:

$$\text{Fidelity} = \alpha + \beta_{\text{QD}}\,\text{QD} + \beta_{\text{EoF}}\,\text{EoF} \tag{13}$$

This predictive relationship demonstrates that QD and EoF together serve as robust, real-time indicators of channel noise for adaptive protocols: enabling our protocol to dynamically optimize purification parameters even as noise fluctuates across distributed links.

As there are no comparable global adaptive purification schemes in existing superdense coding or DQN protocols, this result underscores the novelty and practical value of our approach as a testbed for distributed quantum error correction and hybrid quantum-classical control.

Terms	Value
Mean Squared Error (MSE)	0.000574
R^2 Score	0.97787
β_{QD}	-0.23512
β_{EoF}	0.76228
α (Intercept)	0.47453

Table I: Fidelity predicted from QD and EoF.

V. Adaptive Purification

While the DEJMPS protocol is a foundational method for distilling high-fidelity Bell pairs from noisy entanglement, it exhibits notable limitations in the context of distributed quantum networks:

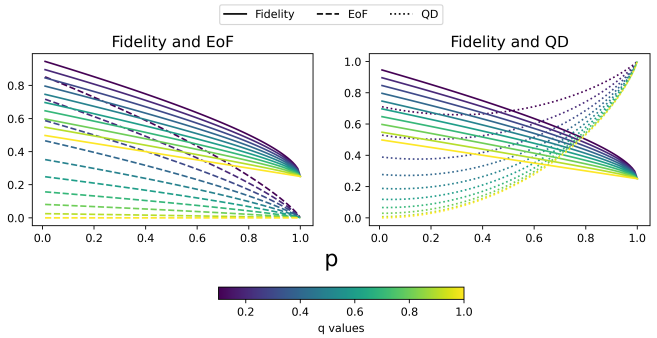


Figure 2: Fidelity, QD and EoF. We have simulated for combined phase and amplitude damping. The x-axis represents p, the parameter for amplitude damping. Color represents q, the parameter for phase damping. Fidelity, EoF, and QD have been plotted for (p, q) pairs.

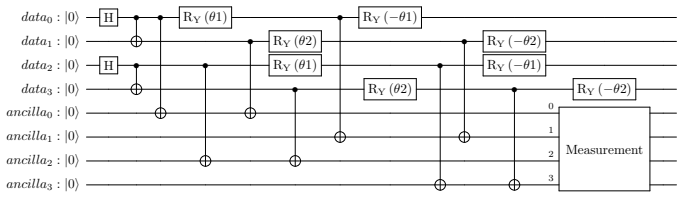


Figure 3: Adaptive purification circuit for distributed quantum networks. Ancilla-assisted unitary $U(\theta_1, \theta_2)$ enables pair-preserving purification across noisy links. Tunable rotations allow adaptation to asymmetric noise; ancillas ensure CPTP mapping for error-corrected, hybrid architectures.

1) Each round sacrifices one entangled pair to purify another, reducing total throughput: a significant bottleneck for multi-user, high-rate quantum networks.
2) DEJMPS is based on LOCC, making it suboptimal for co-located nodes where joint, non-local operations and hybrid quantum-classical feedback are feasible.

To address these limitations, we propose an adaptive purification protocol tailored for distributed quantum networks and testbed benchmarking, with the following key innovations:
Pair-Preserving Global Operations: Joint unitary operations across all data and ancilla qubits allow us to purify entanglement without pair loss, directly overcoming the throughput constraints of conventional distillation.
Noise-Adaptive Flexibility: Tunable rotation parameters (θ_1, θ_2) enable real-time adaptation to asymmetric or time-varying noise across network nodes, a necessity for robust operation in practical DQNs.

Fig. 3 shows the circuit for the adaptive purification module. After decoding, the process can be described as:

$$\rho = \sum_k (\mathcal{E}_k \otimes I)(U_{ij}|\Phi^+\rangle\langle\Phi^+|)(\mathcal{E}_k^\dagger \otimes I) \tag{14}$$

where $\mathcal{E}$ encapsulates channel noise and all operations up to decoding. Direct unitary inversion is generally impossible, so we append ancilla qubits to construct a completely positive trace-preserving (CPTP) map. Specifically, after introducing $|0\rangle^{\otimes r}$ ancillas and applying the adaptive purification $U(\theta_1, \theta_2)$,

the effective action on the data is:

$$\rho' = \mathrm{Tr}_R \left(U \left(\rho \otimes (|0\rangle\langle 0|)^{\otimes r} \right) U^\dagger \right) = \zeta(\rho) \tag{15}$$

The target is to engineer ζ such that $\zeta \circ (\mathcal{E} \otimes I) \approx \mathrm{I}$, effectively counteracting the noise. Unlike DEJMPS, our protocol is not constrained by pair sacrifice or LOCC: the global, ancilla-assisted unitary and feedback-driven control loop enable deterministic, noise-aware correction, well-suited for distributed architectures and testbed experimentation.

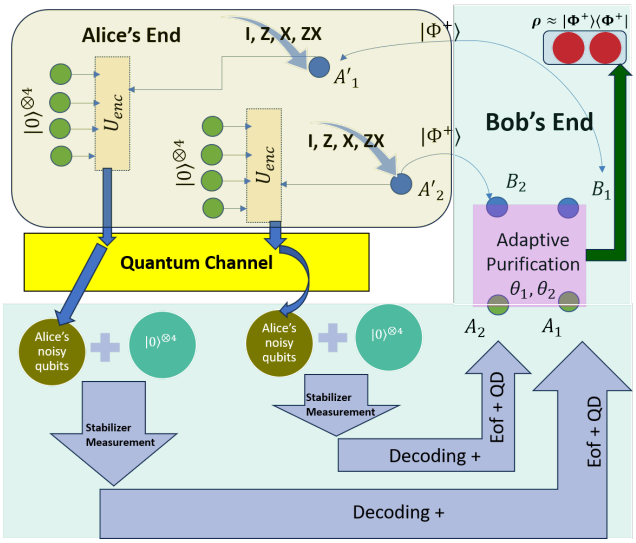


Figure 4: Adaptive superdense coding architecture for distributed quantum networks. Alice encodes data with local operations and a five-qubit code; data and pilot pairs traverse the noisy channel. Bob performs error correction first. Then an adaptive purification based on calculated QD and EoF from pilot pairs are performed.

VI. Proposed Architecture

We consider a distributed quantum network where Alice and Bob share multiple Bell pairs over a noisy link. A subset enables superdense coding, while dedicated pilot pairs monitor the channel environment in real time. This architecture supports adaptive quantum protocols, ensuring robust high-capacity communication across diverse topologies.

1) **Encoding and Transmission:** As depicted in Fig. 4, Alice encodes two classical bits onto each data-carrying qubit (U_{ij}), applies the five-qubit perfect code (U_{enc}) for error protection, and sends the logical qubits alongside pilot pairs through the network channel. The complete quantum circuit for this stage, including encoding, error correction, and decoding, is shown in Fig. 5.
2) **Stabilizer Measurement and Decoding:** At the receiver, Bob uses ancillary qubits to perform stabilizer measurements and error correction, recovering the logical state from the five-qubit code (see Fig. 5). Some residual noise may remain after this process.
3) **Pilot Pairs and Noise Estimation:** Pilot pairs traverse the same noisy channel but do not carry user data. By measuring QD and EoF on these pairs, Bob obtains real-time, link-specific noise diagnostics, enabling adaptive response to dynamic network conditions.
4) **Adaptive Purification:** Based on QD/EoF feedback, Bob applies global, parameter-tuned purification unitary ($U(\theta_1, \theta_2)$; see Fig. 3) to all qubits. This step preserves all entangled pairs, which is important for distributed networks where resource efficiency, throughput matter.
5) **High-Fidelity Outcome:** Bob recovers a set of high-fidelity Bell pairs, supporting robust superdense coding or other quantum networking applications. The architecture enables real-time benchmarking and optimization for distributed quantum networks with fluctuating noise and multi-user connectivity.

Note, the number of pilot pairs used for channel monitoring is a fixed, small overhead, chosen independently of the data payload and amortized over large batches, so throughput is not diminished as the network scales.

VII. Analysis and Discussion

Higher-order and correlated errors remain a fundamental challenge in distributed quantum networks, as standard quantum error correction (QEC) cannot fully eliminate these effects, especially in noisy, long-distance, or multi-user channels. To address these residual errors, we introduce an adaptive purification protocol for distributed settings. Here, "purification" refers to a deterministic, pair-preserving correction, not entanglement distillation, which discards pairs and limits throughput. Our approach thus maximizes resource efficiency and channel capacity while reclaiming the original meaning of the term "entanglement purification."

A central innovation is our use of quantum discord (QD) and entanglement of formation (EoF) as real-time signatures of the channel's noise. Our regression analysis ($R^2 \approx 0.98$, Table I) shows that the combination of QD and EoF offers a much more sensitive and predictive indicator of the underlying noise than fidelity alone can provide. This allows our protocol to adaptively tune the global purification even in regimes where fidelity or Entanglement is low, but useful non-classical correlations remain detectable.

Importantly, the number of pilot pairs required for noise monitoring is constant, independent of the total number of data pairs. This ensures that the relative overhead becomes negligible as the system scales, particularly in multi-hop or repeater-based architectures. Pilot pairs act as probes, capturing the noise signature experienced by all data-carrying qubits, and ensure the adaptive purification remains robust and accurate under a variety of network conditions.

Our simulation framework also serves as a practical testbed, enabling systematic benchmarking of the protocol across realistic noise models such as amplitude and phase damping. By providing this platform as open-source, accessible without specialized hardware, we aim to accelerate the development and comparison of adaptive error mitigation strategies for distributed quantum networks. This supports reproducibility, community-driven innovation, and the advancement of hybrid quantum-classical protocols for future large-scale quantum systems.

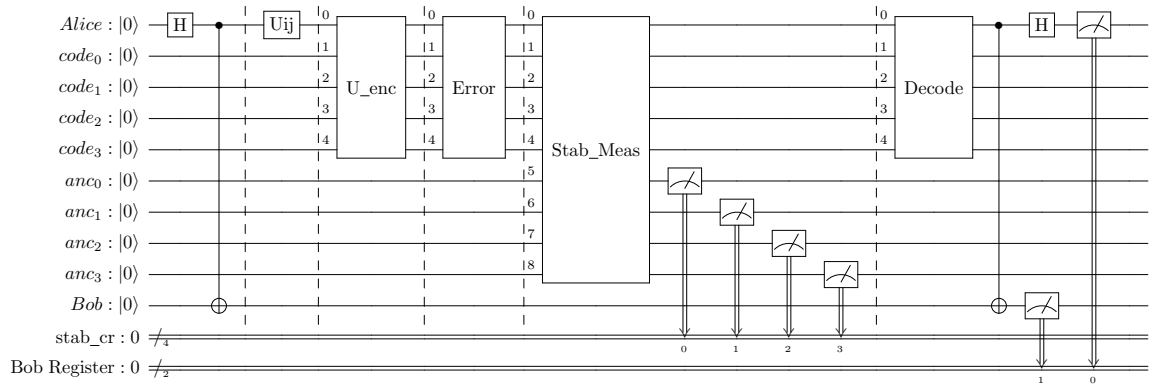


Figure 5: Architecture prior to adaptive purification in a distributed quantum network. Alice and Bob share a Bell pair; Alice encodes and applies five-qubit QEC before sending through a noisy channel. Bob performs stabilizer measurements and decoding, preparing the state for adaptive purification and testbed benchmarking.

VIII. Conclusion & Future Work

We introduce an adaptive superdense coding protocol for distributed quantum networks, integrating five-qubit error correction with global, non-discarding purification tuned by quantum discord (QD) and entanglement of formation (EoF). Proof-of-concept simulations demonstrate enhanced fidelity and capacity under realistic noise, outperforming conventional strategies. Future work will focus on comprehensive end-to-end simulations to refine the mapping from QD and EoF to purification parameters, and to assess the protocol's ability to invert a wide range of noise models. We also plan to incorporate deep learning for real-time parameter optimization in large-scale systems, supported by high-performance computing resources for data generation. Further extensions will include adapting the protocol to depolarizing, thermal, and non-Markovian noise, as well as experimental demonstrations on multi-node testbeds. In summary, this provides a flexible foundation for robust, scalable quantum communication in dynamic multi-user networks, advancing next-generation protocols through pair-preserving, correlation-guided purification.

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